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"Learning Circles: An Adaptive Strategy to Support Food Sovereignty Among First Nations Communities in Canada"

Louise W. McEachern[1*], Jennifer Yessis[1], Barbara Zupko[1], Joanne Yovanovich[2], Renata Valaitis[1], Rhona M. Hanning[1]

[1] School of Public Health Sciences, University of Waterloo, Ontario, Canada, N2L 3G1

[2] School District No. 50, Haida Gwaii, B.C.

*** Correspondence:**

Corresponding Author

lmceach4@uwo.ca

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Abstract

Indigenous communities in Canada are concerned about the health of their youth and the reclamation of traditional food-related skills among their people. Food sovereignty has an integral role in food and nutrition security, and the path to Indigenous self-determination. Learning Circles: Local Healthy Food to School was a community engagement model that aimed to enhance access to local, healthy and traditional foods for youth. In each of four First Nations communities, a Learning Circle Evaluation Facilitator worked to plan and implement activities, build on community strengths and promote partnerships. This paper describes how the model was perceived to support food sovereignty. Data included interviews, process reporting and school surveys, and was analysed according to pillars effective for the development of food sovereignty in Indigenous communities. Goals set by two communities incorporated food sovereignty principles, and in each community capacity-building work furthered the development of a more autonomous food system. There were many examples of a transition to greater food sovereignty, local food production and consumption. Indigenous governance was an important theme and was influential in a community's success. The model appears to be an adaptable strategy to support the development of food sovereignty in First Nations communities.

Novelty Bullets

LC:LHF2S was a community engagement model that aimed to enhance access to local, healthy and traditional foods for youth.

The model is an adaptable strategy to support the development of food sovereignty in First Nations communities.

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There were many examples of a transition to greater food sovereignty, local food production and consumption.

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Introduction

> "*When we strengthen our connection to food, it brings back the traditions – so many of the traditions and the knowledge that has been lost, so food sovereignty is really about revitalising culture, and it's connecting to the land, which is the essence of any culture*" (LC Participant, 2018)

'Health', and health promotion for Indigenous communities is based on the concept of wellness, a balance between physical, spiritual, emotional and mental health (Battiste, 2011; Cochran et al., 2008; Edge & McCallum, 2006; Getty, 2010) in contrast to the traditional focus of Western science on physical health. For Indigenous peoples, food and food knowledge and culture are deeply connected to a holistic view of health and wellness. Rather than be identified by deficits imposed by colonization, such as the sustained relative socioeconomic disadvantage that places Indigenous populations at risk of food insecurity and chronic disease (Kuhnlein et al., 2001; Pioro et al., 1996; Taylor et al., 2005), diverse First Nations, Métis and Inuit groups within Canada choose to be recognized for their myriad strengths (FAO, 2006)(Cidro, Adekunle, Peters, & Martens, 2015; Power, 2008; Reading & Wien, 2009; Willows, N. D., 2005). *Food sovereignty* is a concept linked to the autonomy and sovereignty of a people and one that considers the cultural, political and environmental aspects of food systems (Masioli & Nicholson, 2010). Food sovereignty has been defined by Food Secure Canada as '…the right of peoples to healthy and culturally appropriate food produced through ecologically sound and sustainable methods, and their right to define their own food and agriculture systems" (Peoples Food Policy, 2011).

Food sovereignty is of growing importance to Indigenous communities in Canada and there have been increased references to it over the past numbers of years in published literature (Delormier & Marquis, 2019; Wilson et al., 2020; Wittman et al., 2010). It has an integral role, not only in food and nutrition security among Indigenous populations, but as a path to self-determination for

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Indigenous peoples and non-Indigenous Canadians' reconciliation with Indigenous peoples and communities. Food sovereignty recognises both the relationships and the power inherent in food systems and aims to link production to consumption (Desmarais, 2008; Wittman et al., 2010). A return to a culture of healthy local and traditional foods is fundamental to greater food sovereignty and the promotion of wellness in Indigenous and remote school communities (Battiste, 2011; Kawagley, 2001). With this in mind, and following the First Nation principles of ownership, control, access and possession (Schnarch, 2004), the Learning Circles: Local Healthy Food to School (LC:LHF2S) initiative aimed to increase access to local and traditional food, and in doing so, enhance health and wellbeing in partnering Indigenous school communities.

The Learning Circle (LC) approach is a community-based participatory engagement practice with a collective aim to promote partnerships among community members with a common interest in food. The model stemmed from the 'learning labs' of the Farm to School programmes that began in the United States and transitioned to Canada under Farm to Cafeteria Canada (F2CC) (Farm to Cafeteria Canada, 2019). The approach began as a strategy to increase availability of local healthy food in schools. Establishment of the model as a 'pilot' project in Haida Gwaii (HG), British Columbia was promising (Farm to Cafeteria Canada, Nourishing School Communities & SPARC BC, 2016), hence, the current project sought to explore the sustainability of the model and potential for growth across different local food systems. Upon receipt of funding from the Canadian Institutes of Health Research (CIHR), three other diverse First Nations communities: Hazelton/Upper Skeena (HZ), B.C., Ministikwan Lake (MK), SK, and Black River (BR), MB were invited to participate in a scale-up of the model.

This paper highlights food-related transitions within the diverse First Nations school communities who had participated in the Learning Circle (LC): Local Healthy Food to School (LHF2S) initiative between January 2016 and December 2018; long enough to see their LC goals take effect in local

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food systems. Specifically, we identify for each context, the Learning Circle goals, process and activities. Thematic analysis describes influences, processes and outcomes related to the implementation of the Learning Circle model and transition of local food systems under the project. Findings are then deductively mapped onto six practical 'domains' of food sovereignty (Ray et al., 2019) to demonstrate how the LC model might contribute to food sovereignty in similar communities.

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Methods

Study Design

This community-based participatory research (CBPR) (Holkup et al., 2004) was conducted using a range of approaches to describe the context, process, activities, perceived influences and outcomes of the LC model and transitions in local food systems over 2016-2018. With this approach, the strengths, resources and knowledge of the community are considered foundational and the community is consulted regarding what they perceive as successes or barriers (Holkup et al., 2004) (Jacquez et al., 2013). The context of each community, including location, past community food system projects, and governance, was described elsewhere (McEachern, 2021).

The Learning Circle

The 'Learning Circle' (LC) community engagement practice aimed to promote relationships between members of the community that had a shared interest in increasing the availability, acceptability and consumption of local, healthy and traditional foods by school-aged youth and adolescents. Adapted from 'learning labs' in the United States (Farm to Cafeteria Canada, 2019) the ‘Learning Circle’ name was adopted as the model evolved within HG to reflect the co-operative nature of the practice, and the importance of grounding the practice in local knowledge, traditions and culture, and was preferred by Indigenous communities.

A Learning Circle Evaluation Facilitator (LCEF) was hired by each participating community to initiate LC meetings and provide ongoing support. Community members with an interest in local food were invited through formal and informal networks, and partnerships with key community organisations were created. Interests for each community were discussed and priorities and goals were agreed upon over the course of a workshop, usually day-long. The LC of most of the

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community met several times between 2016 - 2019 to re-evaluate the aims of the group and discuss progress.

A local LC Council was made up of individuals representing various community organisations such as community health agencies, government, and schools, as well as youth and other local interested individuals. Each LC Council was responsible for hiring a suitable evaluation facilitator from the community and maintaining the social and cultural integrity of the LC:LHF2S project for the duration of funding, as well as supporting communities to obtain approvals, implement activities and gather material for evaluation.

Data Sources and Data Collection

Evaluation methods were guided by community priorities, and as a result, multiple sources of evidence were analysed to describe the context and the story of change resulting from the work of the LC. A semi-structured interview script was used to conduct annual interviews with LCEFs, key community members and project partners (n=37). Participants were recruited by email or in person, and were purposively selected based on their connection with the initiative (for example, attendance at the annual project gathering). Interviews (n=52) took place in person, by phone, and on Skype, lasted between 25 – 60 minutes, and were carried out by the same interviewer (LWM) or, in the case of interviews with community members, a trained, local research assistant. All interviews were recorded with the permission of the interviewee and signed and/or verbal consent was obtained prior to the interview taking place. Participants could decline to answer any question that they wished and had the opportunity to withdraw from the interview at any time.

Written documentation included emails, and notes taken during conference calls between project partners, which were exchanged and took place throughout the duration of the LC:LHF2S initiative. After each LC, the LCEF wrote a report describing the participants, events and action plans produced during the meeting. Activity and other tracking data, food procurement data (including receipts),

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menu plans, timesheets and journals from LCEFs, teleconference minutes and descriptions of workshop/food skills class were collected throughout the initiative and emailed to the research team at the University of Waterloo.

Digital photographs and videos were taken by LCEFs, community members, and the research team at each of the four communities throughout the initiative. A school-based student survey, including open-ended questions specific to local food and food-related initiatives and planning, was completed by students in HG and MK.

Data Analysis

All interviews were audio-recorded, and audio files were transcribed verbatim; pseudonyms were assigned to each participant to protect anonymity. All other written documentation, including photographs, were reviewed and coded using the following steps. A draft coding framework was developed using activity tracking documents and early participant interviews. These codes were established using both an inductive approach (observing themes emerging from the data) and a deductive approach (exploring the data for themes relating to Indigenous food sovereignty (Ray et al., 2019)); the codebook was verified in discussion with the project research team. Due to the COVID-19 pandemic, the main themes emerging from the analysis were discussed with community advisors remotely, as part of a 'sense-making' session. Data were then coded line-by-line and analysed for emerging themes by comparison across the documentation, and within and across communities. Reports of LCEF activities were coded and summarised by one coder (BZ), and coding of interviews was carried out in duplicate (LWM and one other coder). NVivo® 12 Pro (QSR International) was used for data processing and analysis. Methodological rigour was supported by member checking, and the large variety of data sources included in this study allowed for triangulation of data, contributing to methodological rigour(Creswell & Miller, 2000; Creswell &

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Poth, 2017; Stewart et al., 2008). Due to the COVID-19 pandemic, main themes were discussed with the communities remotely, as part of sense-making of the themes that emerged during analysis.

A review of conceptual frameworks linking Indigenous food sovereignty to health identified six practical 'domains', recognised by others as being important for effective development of food sovereignty in Indigenous communities (Ray et al., 2019). These domains are listed below:

- Policies and procedures
- Local food production and consumption & Local/cultural knowledge and practice
- Self-determination and governance
- Social justice and equity
- Health of the land
- Adaptability and resiliency

Findings were mapped deductively onto these domains to illustrate how the LC:LHF2S model might support food sovereignty in Indigenous communities.

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Ethics

Ethical clearance was received from the Office of Research Ethics at the University of Waterloo, (ORE# 30819). An initial partnership and collaboration document, written in plain language, was given by research partners to the LCEF of each community to share as appropriate with community leadership. This document covered how OCAP was integrated into the project and plans for knowledge sharing. In 2018, an agreement, '*Isda ad dii gii isda (S) - Isdaa 'sgyaan diiga isdii (M)*' Spirit of Collaboration Protocol was established with the University of Waterloo and the Secretariat of the Haida Nation (SHN); for the other communities, the management and oversight of evaluation activities was established in a less formal way. OCAP and principles of respect and reciprocity were carried out in this research by adherence to the local decisions of each community regarding what

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information was desired, and which processes were used for collecting information (Willows, 2013). Members of the local LC Council made the final decisions regarding which methods were used to evalute the LC in each community. Two communities chose to hire their own research assistants to conduct interviews with community members, and each community was given access to the interview transcripts relating to them. The LC Councils were responsible for releasing data to the research team, and were consulted during data analysis.

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Results

Context

The story of the LC:LHF2S in each of four distinct FN contexts was varied, though common themes and influences were identified. As the 'pilot' community, the LC in HG had started in 2013. As part of the formative work for this, CIHR-funded, study, a Haida community member was hired to conduct interviews with community members relating to the existing LC. Haida community members who were interviewed noted that, although they benefited in some ways, they did not actively participate in the LC or its activities, and that the LC was not principally their initiative. HG has a rich food environment, due, in part to its temperate climate, a vibrant local and traditional food culture and a long history of protecting the land and culture. A variety of local food-related activities had been taking place in HG prior to the establishment of the LC project in 2013, and by 2016, the LC work was firmly established and included the new initiative of the local food pantries. Under the CIHR funding from 2016 - 2019, the LC activities in HG focused on three main areas: schools, hospitals, and two local food pantries. Local food pantries aimed to increase year-round access to local foods by providing a point of focus where food is purchased or donated, processed and stored for distribution to schools and other participating public organisations. Within the overall work, LC participants worked towards increasing access to local (i.e., garden produce) and traditional (i.e., foods of cultural significance, including salmon, berries and sea asparagus) foods, building knowledge and skills, fostering relationships, and transitioning to Haida leadership within the project.

While the HZ region had some local food activities prior to 2016, there had been no organised community-wide commitment to the food culture and environment and so the LC initiative in this community focused on partnership development, gardens, community-wide skills work, and youth activities, with a specific emphasis on schools. Foods traditionally eaten in this region include salmon, oolichan, wild meats (such as moose), berries, vegetables such as root vegetables, and

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traditional medicinal plants. Traditional food, knowledge and practices were prioritised in youth-based programmes, for example, the 'Brighter Futures' programme that organised youth trips onto the land for berry and medicine picking in a respectful way, as guided by Elders or Knowledge Keepers. The LC in this community received administrative and advisory support from the Storytellers' Foundation; the organisation was responsible for advertising for and hiring the LCEF.

The MK community in Saskatchewan is a small, remote community located an hour's drive from the nearest well-stocked grocery store. The community had a long history of gardening, but many traditional skills were lost due to the impacts of colonization; in the time of the Indian Agent (prior to the 1960's) government rations were based on a perception of how much food a community had access to, and fewer rations were given to families that grew their own food (Irwin, 2018). Some local food-based activities were happening prior to 2016 but there had been no organised efforts towards local and healthy food work. The LC initiative focused on capacity building, gardening activities and the school lunch programme. Cultivating youth leadership was prioritised here, and connections between the youth and Elders were developed; the importance of local champions and challenges related to reliance of local champions were noted. The LC in this community received administrative and advisory support from the MLTC; the organisation was responsible for advertising for and hiring the LCEF.

BR, MB, had prior involvement with school health programming (Abramowicz, A., Yessis, J., 2017), and were invited to participate in the LC initiative by virtue of prior partnerships with the research team. They joined the LC initiative in 2017 and while they had not had an "official" LC with others in the community at the time of evaluation, local food activities in BR centred around the school. Activities involving youth included berry picking, wild rice harvesting and fishing.

The communities are compared by context in Table 1. The population size of HG and HZ were similar, at approximately 5,000 people, and the population sizes of MK and BR were also similar,

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ranging from 1,000 to 1,300 people. HG and HZ have six and seven schools respectively, and MK and BR each have one school. With the exception of HZ, the administrative agreements made between the University of Waterloo and the community were with Indigenous leadership organisations.

Goals

The goals articulated by the LC in each community are compared in Table 2. Three communities: HG, HZ, and MK, recognised the need to obtain supplementary funding in order to support the action plans prioritised by the LC. This funding was used to purchase school food and kitchen equipment such as refrigerators and freezers, and also for gardening equipment such as materials for raised beds, tower gardens and greenhouses. Funding was received from local governance (Northern Health; Meadow Lake Tribal Council) and NGOs (F2CC, Mazon Canada, Whole Kids Foundation).

Increased access to local and traditional foods for school children was a focus for HG, HZ and MK, and increasing connections between Elders and youth was specifically articulated by HG, but also a part of activities in HZ, MK and BR. The local food pantries in HG provided activities that incorporated local foods into the school curriculum, and skills classes were built into the curriculum by a number of teachers. Curriculum development was identified as a specific goal by HG and HZ, and carried out by BR.

The development of partnerships was a specific goal of HG and MK, and although not identified as a goal, was a large focus of the activities of HZ. The purpose of this in all communities was to extend the reach of the LC and make it a 'whole community' initiative, include as many knowledge-holders as possible, build economic relationships with growers, share resources (including food plus learning), build capacity, and further establish efforts to build food security and food sovereignty.

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Black River did not articulate any goals, however their activities focused on developing connections between Elders and youth, traditional foods and skills, and curriculum development.

Activities and Themes

The activities carried out by each community are compared in Table 3, along with the main themes emerging from the work in each community.

Activities

Skills development on the land with Elders, and capacity building was a focus of the LC activities in all four communities. This included activities as disparate as grant-writing workshops in MK, school menu planning by the community dietitians in HG and MK, to children's cooking classes in HZ and HG; learning activities with Elders were carried out in all four communities. Gardening activities were carried out by all four communities and mostly took place in schools; HZ and MK also had some community gardens that were a focus of the LC work. Gardens were used as a tool for engaging youth, and, in HG and MK, the food produced in the gardens was incorporated into the school lunch menu. A 'healing garden' was developed as a mental health support for youth in HZ. The provision of food for schools was a focus of HG, MK and BR, and was offered as either a breakfast (some schools in HG, BR) or a lunch programme (MK, some schools in HG). Developing youth leadership skills was a specific aim and focus of the LC in MK; youth were included as members of the school council in the LC meetings and were asked to provide feedback as part of a school survey. However, contributions to youth development were included in all four communities. For example, youth were a focus of leadership activity workshops in HZ (i.e., 'Youth On Water' leadership programme) and in BR the Cultural Exploration School Initiated Course helped to develop youth skills and encourage the development of youth-Elder relationships.

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The provision of food for hospital meals was a focus of the LC in HG only, as was the development of the local food pantry within this community. HG was the exemplar community and as such built upon a strong foundation. Many of the activities that had begun prior to the current evaluation were given time to establish themselves in the community and continued to flourish.

In addition to the activities listed in Table 3, all four communities engaged in cross-community collaborations to learn from one another during the initiative.

Inductive Themes

Increasing access to healthy and local foods within the community was a strong theme that came through in the documentation and interviews of each community, for example, school food programmes and the focus on gardening. The local champion was also an important theme – notable as much by the absence of a champion as the presence; multiple committed champions were key to the work in HG, HZ and MK, but within the small community of BR, capacity to champion new initiatives was stretched and could have contributed could have contributed to the poorer uptake of the LC model.

> *"...in Ministikwan, she has been, I think the greatest help with adapting people. Um, the health clinic is kind of the communication hub of the community in Ministikwan. And..she's a health worker there and um, she's always willing to help people and bring people together for different healthful reasons. Um, and she's been indispensable, I will say um, in many endeavours, I think she, she's a champion of her community."* (Community Dietitian, MK, 2017)

Building partnerships and community togetherness was seen as important in all four communities.

> "*We all have an interest in local and traditional food...We all like to laugh, we all love cooking, eating and sharing*". (LC Participant)

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> *"[the LC] is building relationships in the community and it is connecting places like Senden to the elementary school, which is a really beneficial relationship. It helps both programmes sustain themselves, so I think those things are really great successes that we have seen. I know my own kids being in the school get really excited. You know they brought home produce last year and were super excited that it was from their garden boxes*." (LC Participant)

> "*I really feel like, with reconciliation, there is work that both Indigenous and non-Indigenous people need to do. And some of that work non-Indigenous people need to do is hard and it's uncomfortable*" (LC Participant)

In addition, increasing access to traditional food, and building traditional knowledge and practices was also critical.

> *"The hospital is starting to serve traditional foods. Before they were serving foods that came all the way from Toronto in packages. We didn't like it when food was coming all the way from Toronto. Traditional foods are healing for our people."* (Community Elder, HG 2018)

> "*[I am] proud of the work we are doing at the school, making healthy food 'cool', [and] working for pride in the culture*" (LC Participant, HG, 2018)

> *"One of our goals for today is to review past goals and visions – are they still relevant? Are local foods cost efficient, enjoyed, and easy to use…Do they strengthen traditions and connection to land/sea, and can we figure out ways to incorporate them into curriculum?"* (LC Participant – as part of an introduction to the 2017 LC)

> "*My experience working with children and families and schools is to get people outside of the classroom. So, getting them out on the land or out on the water with our people. And that's where our people shine, that's where we're the most confident, where we're the most alive is*

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328 *when we are out on the land. So, if we can get our teachers and our kids outside. I think that's*
329 *the most beneficial thing we could do. I don't know how much of that has happened over the*
330 *last two years*." (LC Participant, HZ 2017)

331 Enhancing Indigenous leadership within LC:LHF2S was a goal of the LC scale-up in HG, and
332 involved the formation of the Haida Foods Committee and their leadership of the LC and research
333 oversight.

334 *"Nothing happens in a void – the fight for lands has been important and is ongoing; title*
335 *case, land stewardship, Haida language – it all contributes to taking ownership of our own*
336 *health."* (LCEF, HG)

337 Indigenous leadership was also integral to the LC in MK where there were very few non-Indigenous
338 community members.

339 *"I think it's really important how you package it. We could package it in a way where we*
340 *create champions and foodies in the communities, whether it's externally or internally, with*
341 *the Indigenous people, or we can really focus it and sell it in a way where the community*
342 *becomes more sovereign in their thought process and their actions, and they actually own*
343 *this, and this is for them, by them, with them, and really taking control of their health, their*
344 *lives, their quality of life. It was something that was done generations ago. Our ancestors did*
345 *it all along."* (LC Partner 2018)

346 Despite Indigenous leadership and support for the initiative in BR, no specific LC was held. The
347 lack of Indigenous leadership of the LC model was a challenge in HZ, and leadership of the initiative
348 in HZ by a local non-Indigenous NGO may have contributed to the lack of involvement of some
349 schools on reserve lands.

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Deductive Themes

1. Policies and Procedures

The development of national-level policies such as school food policies are necessary, but beyond the scope of the LC model. However, in communities that had the requisite capacity, policies and procedures at the community level, for example, incorporating traditional knowledge into the school programme such as including butchering workshops, were developed by the participants connected with the LC.

Refocusing the school curriculum towards traditional practices was an aim of the LC in HG and HZ. Changes to school curricula relating to traditional practices and foods were written into the goals of the LC in HG and HZ with the aim of developing traditional food literacy through practical and theoretical ways. Goals from HZ included:

- Value local food (and work to increase its value among community members)
- Connect kids with a healthy food culture on the territories.

Goals from HG included:

- Support access to Haida traditional and locally farmed foods through greater coordination, communication and culturally appropriate ways of growing, harvesting, preparing and trading among schools and beyond.
- Promote youth engagement and leadership in LC:LF2S programming; bring youth and Elders together to support knowledge sharing around food and stories

In HG, skills classes were carried out in school including those led by local Elders, and a seasonal calendar was developed with local Elders including the Haida names of foods and what and where to gather each month. In HZ, traditional foods were incorporated into art classes, and skills were

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learned in the school gardens. In addition, organisations that partnered with schools in HZ, such as Senden Sustainable Agricultural Resource Centre, were instrumental in facilitating hands-on learning and adding to what adolescents were taught in the classroom. The LCs in MK and BR did not include curriculum changes in their goals, but they did carry out LC activities among their students. In BR, the Cultural Exploration School Initiated Course, a provincially supported programme that recognises curricula developed by schools and school divisions to meet local needs and interests, provided an opportunity for students to learn more about their traditional heritage and culture while at the same time earning high school credits. Each high school student participant was required to spend 110 hours with local Elders participating in traditional activities to learn more about their heritage and culture. In MK, a seasonal calendar was developed between youth and Elders during school hours and traditional foods skills, such as butchery, were demonstrated in school workshops. In addition to adjustments to school curricula, adherence to traditional protocols and the use of appropriate language when talking about and referring to food was an important component of the learning circle work in all four communities.

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At the school board level, the School District responsible for schools in HG (SD #50) had nutrition interventions written into the outputs and outcomes section of its 2011 logic model. In HG processes have been developed to safely include game meats in school and hospital meals (Logan, 2021). However, food safety policies were seen as a barrier to incorporating traditional foods into school meals in HZ, MK and BR. Food safety legislation relating to the consumption of hunted meats in schools was the main challenge.

2. Local Food Production and Consumption, and Local/Cultural Knowledge and Practice

Increasing food access and changing the food environment was a theme of the LC-related activity in all communities, and each community included activities that contributed to the production and consumption of local foods, focusing on youth, the connections between youth and Elders, and in

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HG, hospital food. Hand-in-hand, for the communities studied here, were activities relating to traditional food knowledge and practice.

Work among the community schools was a major focus of the work carried out by the LCs relating to foods and cultural knowledge. In HG, the pantry supplied foods for cultural camps and many pantry workshops had student involvement. MK and BR had daily school lunch or breakfast programmes, respectively, and MK focused on school and community gardens, aiming to use skills within the community, for example, with permaculture. In HZ, school lunch programmes took place in Majagaleehl Gali Aks School and NH Elementary School, and all three schools connected to the LC initiative introduced gardening and skills workshops for their students.

LC goals in HZ and HG pointed very much towards food sovereignty (e.g., 'value local food and work to increase its value among community members'; 'connect kids with a healthy food culture on the territories') and learning traditional skills was central to the work of their learning circle.

> *"I think it's a combination of the two. You can't talk about local food, without talking about Gitksan traditional food systems. That's just part of who we are…. We also have [teacher] she comes in and if we get fish or we get oolichan, or even get bear or whatever. She'll come into the class then we learn how to either smoke it or process it or whatever. Bear was a really interesting one because in our community people from different house groups and different families have different beliefs on bear meat. So, some families say that's totally fine, in other families it's a sacred animal. So, it became a really interesting place for conversation about tradition and about culture and how it can be different even within the same nations."* (LC Participant, HZ, 2018)

> *"I'm also lucky to have …. Our First Nations Resource Worker, and she's been very active with coordinating… going berry picking with kids; or we had another Haida role model come and teach our kids how to butcher and deal with some beef – some local had donated a cow –*

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and canning salmon, and just some of those skills that some of our kids have active parents and family members that help teach them, but not all of them." (LC Participant, HG, 2017)

Four partner organisations in HZ were connected with food-based projects and all of the schools had gardens in which students were able to learn skills; partner organisations such as Senden provided skills training, and the Storytellers' Foundation provided workshops teaching food preparation and preservation skills to local children.

In HG, the local food pantries were the main hub of the production activities as they served as a central point for the collection (purchased or donated), processing and distribution of local foods. Pantry staff also organised field trips, workshops and garden activities that were attended by school children, parents and other community members. The teaching at these events was often carried out by Elders – serving the dual purpose of connecting generations and providing valuable traditional skills. HG also had a hospital food programme, which involved working towards removing the 're-therm' food and including local and traditional food on the menu. The significance to community members went far beyond nutritional value and extended to the inclusion of culture, family and Indigenous ways of well-being into the experiences of those staying in a Western-style hospital.

School lunch programmes, particularly that in MK, were highly significant; students recognised that some of their peers come to school hungry.

"I don't really like nothing about it but I'm glad they do this so that kids that don't bring food don't have to starve." (School student, MK, 2018)

The provision of the food was also seen as an expression of the care of the lunch programme coordinator. Many students noted the quality and healthiness of the food and a number of students appreciated the fact that the food is made and served by students and valued what they learned through the programme.

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"I like that it has many options for students in case they don't like something. It's nutritional and it's also welcome for every student" (School student, HG, 2019)

"*We don't have to pay; our food is free and it's really good*" (School student, MK, 2018)

"I like the idea that we also get to learn how to cook and provide free food to everyone at the same time" (School student, HG, 2019)

Students also reported an interest in consuming more traditional foods, demonstrating the importance of tradition in the lives of these students and providing reassurance for the older people in the communities who are concerned about the loss of culture among youth. When asked what foods they would like to see more of, a number of students reported an interested in seeing traditional foods on the school menu, for example:

"..traditional food or animals like moose deer elk beaver etc." (School student, MK, 2018)

The price of local foods was reported to be a barrier to the consumption of local foods. Logistics, such as the availability of suitable food preparation equipment in schools, was also listed as a barrier.

3. Self-Determination and Governance

One of the HG's goals under the period of the CIHR Pathways funding was to transition to Haida leadership and the development of a memorandum of understanding between the research team and the *Xaayda-Xaada* Foods Committee (XFC) was finalised in 2017.

In MK, the Island Lake School Student Council was invited to the LC meetings and encouraged to become involved in the decision-making process. By doing this, the community hoped to build capacity and foster motivation within the youth themselves by giving them leadership opportunities, and to learn their thoughts regarding the food-related needs within the school. Members of the Student Council were asked for their opinions on the current lunch programme and were influential in the development and implementation of the school survey. Their input was used to inform and

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focus LC activities: for example, they expressed an interest in including more traditional foods in the school lunch programme.

In HZ, inter-community tensions were identified as a legacy of residential schools. However, bridges were built over the course of time between Indigenous and non-Indigenous community members.

The LC model did not gain traction in BR, likely due to limited capacity in this small community for the diverse stakeholder input that LC involved elsewhere.. Within the school, however, they embraced traditional teachings and connection to the land as a means to connecting youth with culture and local food systems.

4. Social Justice and Equity

This domain focuses on ensuring access to traditional territories and the equitable participation and sharing of food system benefits. Ensuring access to traditional lands was not the focus of the LC model, but the initiative took place at the same time as other events such as the Wet'suwet'en demonstrations (Bogart, 2020), which happened close to HZ.

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Opportunities to bring youth out onto the land to learn traditional skills were important, with most of these incorporating opportunities to interact with and learn from Elders. Land-based learning is important as it encompasses culture, family, food and land. Youth were also engaged in local food activities and reported that they took pride in the work they did outside on the land and valued the effort taken in growing and providing food for themselves, their families, and communities. The work gave youth a sense of purpose, and Elders and older adults involved in the learning circle considered it a blessing to be able to do work on behalf of the children.

> *"We are all there for the kids – [we] all live on Haida Gwaii and harvest traditional foods, all looking to gather and share knowledge"* (LC Participant, HG)

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In HZ, a lack of Indigenous engagement with the project was perceived by community members to stem from the LC initiative and project management being external to the Gitksan and colonial associations with agriculture and Western research. As a result, a number of on-reserve schools chose not to engage with the initiative. Similarly, the LC in the exemplar community of HG had been established independent of Haida leadership. The Haida community, while valuing the work, did not, at first, feel it was 'their' initiative.

5. Health of the Land

This domain focuses on protecting the integrity of territories and natural environment and was discussed by a number of community members and project participants in interviews and group meetings. A number of communities were reluctant to share traditional wisdom and especially source locations for traditional plants or animals out of concern that traditional protocols would not be followed, or plants harvested in ways that did not ensure sustainability. The traditional emphasis on sustainability, and the positive impact that traditional ways have on the environment was discussed by a number of LC members.

> *"[Participant] spoke about the value of food in relationship to the land and to being a Gitksan person. She talked about everything having value, that some parts of the salmon considered a delicacy are sometimes wasted by people. All parts that are not used must go back into the river to continue to feed the river. She talked about fish strips feeding children and allowing them to continue playing and learning. She mentioned that it is important to always have enough to share with all the children. "Use everything," she said, "everything has value."* (LC Report, HZ)

6. Adaptability and Resiliency

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A focus on the long-term sustainability of the initiative was a theme seen across HG, HZ and MK. Funding availability for school food programmes was a concern, along with funding for the infrastructure connected with these programmes. However, as result of investment in local foods, communities saw an economic benefit to the community, particularly in HG.

> '*As idyllic as we make Haida Gwaii seem, there is also poverty here and health problems. So, giving back means that you fund the positions that support harvesting from the land, and you bring meaningful work to the community. Poverty and unemployment is less of an issue when you have a robust local food system. The local hunters, farmers and gatherers benefit from this because their time working is valued by our purchasing. These are the ripple effects: local people gathering foods, processing the foods, distributing it on the island, cooking the food. This all weaves together into what is healthy for us and the community.*' (LC Participant, HG, 2018)

Building and enhancing capacity within the communities was also an emphasis of the LC work in all three communities listed above. Additionally, investment in youth contributes to the sustainability of the initiative, as youth began to take ownership, for example, the student leadership council in MK, the gardens as mental health supports in HZ, and the teaching of traditional food skills across programmes.

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Discussion

The multi-faceted potential of the LC:LHF2S initiative was compared with the image of a crab: "*A Crab has many arms, [is] grasping for many ideas and learning many ways*" (LC Participant, HG, 2018), and this study documents the large amount of work carried out by partners within the four Indigenous communities in this project. A LC uses an engaged scholarship or action research approach to involving communities in activities while collecting knowledge, and the findings of this study indicate that a LC is a feasible venture in Indigenous communities and one that can facilitate building of partnerships, foster actions to increase access to local and traditional food among school-aged youth, and enhance food sovereignty.

The results from this study show that LC:LHF2S is a flexible initiative that engages the broader community in planning and implementation to build upon community strengths and consolidate local food efforts, previously happening in an *ad hoc* way within the community, among both Indigenous and non-Indigenous community members. As such, the LC approach reaches beyond individual communities, includes sharing across communities, and incorporates some of the best practice recommendations for community-based initiatives in Indigenous communities (Bhutta, 2017; Ki-moon, 2016; Sheehan et al., 2017). Experience from HG shows that sustainable initiatives can emerge from grassroots movements given enough funding, time, and commitment. Experience from HZ and MK shows that partnership development and relationship building are an essential early component of a successful LC; community togetherness (including Indigenous ownership of the initiative) and multiple local champions are vital, and capacity building is important in smaller communities that face a number of social challenges. Experience from BR shows that strong internal and external support systems are necessary for the LC model to be effective. Indigenous communities face a range of competing priorities and champions can become stretched: despite support for the LC initiative from community leadership the LCEF was called upon to support a

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552 number of other aspects of health and education within the community. While there may be a level
553 of capacity within a community to support a Learning Circle workshop and process, this did not
554 preclude positive local and traditional food actions within school communities.

555 The successful adoption of the LC approach in schools in three communities (HG, MK, BR) supports
556 research that indicates the value that Indigenous communities place on school-based food
557 programmes (Gates et al., 2015; Hanbazaza et al., 2015; Hanning et al., 2011; Skinner et al., 2006).
558 However, the focus on schools in HZ appeared to be a barrier to the success of the LC. None of the
559 on-reserve schools became involved with the LC and there were challenges at times with engaging
560 the wider Indigenous community. It was suggested that aversion to agriculture and gardening in this
561 community is strong, tied to persistent effects of residential school trauma (Angel, 2000; Miller,
562 2019) and colonisation. Using the medium of a Western-style school as the focus of this initiative
563 was also raised as a barrier. In addition, while there was engagement and support from Indigenous
564 leadership within the community, leadership of the LC remained with Storytellers' Foundation (a
565 non-Indigenous NGO) throughout the course of the initiative and this made those on the research
566 team consider whether 'two-eyed seeing' – the lens of both perspectives – was implemented
567 successfully in this community. This contrasts with other communities, for example MK, where the
568 integration of the two-eyed seeing approach was evident; permaculture and aeroponics gardening
569 methodology was added to traditional approaches to planting, akin to the synergy of the three sisters,
570 and ceremonial tobacco. In HG too, two-eyed seeing was seen through the emphasis on ownership of
571 the initiative by Haida people through the XFC.

572 For Indigenous peoples in remote areas, food system changes have tended to decrease dietary
573 diversity as a result of a lower intake of traditional food and a greater reliance on limited types of
574 market food (Kuhnlein & Receveur, 1996). The goals of the LC initiative included actions that
575 would lead to increased dietary diversity among these school children, and the story of the LC across

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576 these four communities is one that includes changes to the food system, e.g., through changes in
577 school meals to emphasise more local and traditional foods, to changes in student skills in procuring
578 and processing foods from the land. For the three new communities, people involved in the existing
579 food work in each community were brought together and the work was thus strengthened, and
580 important groundwork was carried out with respect to partnership development and relationship
581 building. The LC provided a focus for all of this work, giving an opportunity for ideas to be
582 discussed and connections to be made in order for new food activities to be carried out – i.e., new
583 gardens being developed, distribution of food from local suppliers, sharing of food/traditional skills.
584 For the exemplar community, HG, the LC became less important. The work here had already been
585 established and the further funding of the LCEF through CIHR provided the support to enable the
586 work to diversify into a co-operative style operation – developing into two food hubs (north and
587 south) with different components and multiple champions.

588 The LC approach has been shown to be a feasible and appropriate way of engaging community to
589 support access to local and traditional foods and associated knowledge and practices among
590 Indigenous youth. The LC model could also be a means to support food sovereignty among
591 Indigenous communities in rural and remote locations. The core of the principles of food sovereignty
592 is that Indigenous people should have control over what and how they eat. Although the term, 'food
593 sovereignty' was not language used by participants, it is clear that the components of food
594 sovereignty (Policies and procedures; Local food production and consumption & local/cultural
595 knowledge and practice; Self-determination and governance; Social justice and equity; Health of the
596 land; Adaptability and resiliency) were implicit in the. work that took place under the LC initiative
597 2016 – 2019.

598 A number of qualitative studies have explored the development of food sovereignty among FN
599 communities or community groups as a result of specific initiatives (Hanemaayer, 2019; Schembri,

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2021). However, to our knowledge, this is the first study that has evaluated a scale-able model to support community-led food sovereignty in the Indigenous context (Ray et al., 2019).

Our findings demonstrate that the LC model has impacts on each of the food sovereignty domains developed by Ray et al (Ray et al., 2019). Community advocacy has the potential to influence many aspects of policy and procedure at the community, municipal, and school district level. Changes to school curricula and the incorporation of traditional food and culture into the school day on a regular basis are examples of the work that has been done among the communities studied here. Student voices from the two school surveys conducted in MK and HG tell us that some students in these communities are coming to school hungry, and that this is recognised by their peers. Unfortunately, the literature demonstrates the unplanned impacts that public health programmes and a lack of emphasis of traditional food in Indigenous schools have on Indigenous children (Kuhnlein & Receveur, 1996). Previous work illustrates the desire to consume branded foods along with the association of 'branded' foods with the ideals of 'the good life', and the negative association of traditional foods as being 'backward' and un-modern (Companion, 2013; Damman et al., 2008). Encouragingly, students in HG and MK enjoy eating game, wild fish and locally grown vegetables and students in both communities would eat more traditional foods if they were available at school. This presents a significant opportunity for these communities. Comments expressing the need for the inclusion of traditional ways of knowing into the school curriculum echo similar themes in other studies (Naylor et al., 2010).

Food safety legislation relating to the consumption of hunted meats in schools was a major challenge, although steps taken by the community in HG led to the development of a safe abattoir for the procurement of culled deer meat from Parks Canada (Parks Canada, 2020). The School District responsible for schools in HG (SD #50) had nutrition interventions written into the outputs and outcomes section of its 2011 logic model. While not developed by the LC in HG, this is an example

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of policy work that could be carried out via advocacy by LC members, as it has potential for high impact on community food sovereignty. Other activities critical for working towards food sovereignty include a focus on increasing food access and changing the food environment towards the inclusion of more traditional and cultural food and knowledge activities. LC goals in two communities pointed very much towards food sovereignty (e.g., 'value local food and work to increase its value among community members'; 'connect kids with a healthy food culture on the territories'), and practical activities to work towards these goals were a strong theme of the LCs in all communities. From the perspective of school food services, a barrier to the consumption of local foods was the price – cited as being much higher than non-local equivalent foods – this could affect the success of such food sovereignty efforts among FN communities.

Indigenous leadership and ownership of the initiative was considered essential to the success of the LC across all communities, emphasising the importance of the 'self-detemination and governance' domain. In HG; the development of a memorandum of understanding between the research team and the XFC was very important on a practical level, but also symbolic of a shift in the relationship and ways of working together. Transitioning to greater Haida leadership involved building respectful relationships and trust and adhering to OCAP principles for programmes as well as research.

Long-standing inter-community tensions in HZ may have influenced the commitment of Indigenous community members to LC:LHF2S. Both of these factors may have acted as barriers towards the realisation of Indigenous food sovereignty through the LC initiative in this community. These long-standing tensions, whether through distrust of colonial structures like schools or gardens, external researchers or non-Indigenous project leadership were a challenge at the beginning of the initiative, but with commitment, bridges were built over the course of time between Indigenous and non-Indigenous community members. This is an example of respectful relationships being integral to the development of food sovereignty (Wittman et al., 2010).

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The inherent flexibility of the LC model aligned with Indigenous emphases on relationships, youth, traditional foods, culture and sustainability and where these strengths were emphasised, e.g., with school programming focused on traditional knowledge and skills, and Elder-led community workshops, there were many examples of Indigenous engagement and positive action.

Strengths and limitations

The LC:LHF2S study was primarily a work for and by the communities, and the evaluation was therefore guided by the priorities of the communities. Communities were asked for their definition of success prior to the planning of evaluations to ensure that the information gathered would be as useful as possible to them. This was important from the perspective of OCAP, and also adhering to the spirit of community-based participatory research.

Had the evaluation been planned solely from a Western methodological perspective there might have been comparable data from each community across multiple time points. A strength of the research is that communities identified what evaluation data would be most helpful to them and the processes through which the data were collected. Hence, in addition to LCEF and LC reports from all communities, approaches varied, e.g., two communities opted for student surveys and two conducted interviews of community participants. Data were collected by community members familiar with community, language and culture and research team members informed methods and tools. Distance and this relinquishing of control over process was at times uncomfortable for the the non-Indigenous research team but ultimately was considered an testament to the reciprocal relationships developed.

Implications

Where food insecurity is high, small grass-roots programmes will be insufficient to address it. Nevertheless, the unique context of each Indigenous community and their distinctive needs and interests suggests that local healthy and traditional food initiatives have value and importance.

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Hence, there is still a role for those movements when thinking about long-term food sovereignty and advocating for larger changes, such as national food policies.

Community findings point to strengths, including preference for nutrient-dense traditional foods and limited exposure to "fast" foods and restaurant meals and the importance of school food programmes. Targeting the diets of children/youth via school programmes is a popular approach for enhancing health in the Indigenous communities. Consistent with the need for such programmes to be culturally relevant, (Gillies et al., 2020), the preferences reported by the students surveyed in HG and MK certainly indicate that traditional foods would be welcome as part of school food programmes. In addition, traditional food skills activities are enjoyed by students and add another layer of richness to school-based programming. The focus on youth aligns with the seventh-generation principle – yet schools were also perceived, by some, as colonial structures that may have limited the Indigenous engagement in the current project. Communities and LCs may prefer other ways to engage youth.

It is beyond the scope of this study to examine long-term outcomes of the initiative. In the new communities, participants felt that some relationships and initiatives would continue while others might be more dependent on external funding and the availability of a local champion. The future progress of the work begun here would be an interesting further project.

Conclusion

The formation of strong relationships is an integral part of food sovereignty. New and renewed relationships must be built, not just among people, but between people and the land (Wittman et al., 2010). This theme can be traced through the stories of the communities in this study. Developing trust and community connectedness was critical to the forming of a strong LC partnership, and is integral to many of the domains mentioned above. Strong community champions were important for building relationships and food became the focal point of connection within all of the communities, breaking trails towards reconciliation, acceptance and trust.

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Food sovereignty is of growing importance to Indigenous communities and there have been increased references to it over the past numbers of years in published literature. Food sovereignty has an integral role, not only in food and nutrition security among Indigenous populations, but as a path to Indigenous self-determination and reconciliation by non-Indigenous Canadians towards our First peoples. The LC model appears to be an inclusive and adaptable strategy to support the development of food sovereignty in rural and remote communities.

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718 **Data availability statement:** Data are available upon request, providing permission is granted by
719 the requisite community/communities.

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852 **Tables**

853 Table 1: LC:LHF2S Communities Compared by Context

	Haida Gwaii	Hazelton/Upper Skeena	Ministikwan Lake	Black River
Population	5,000	5,000	1,300 (on-reserve)	1,000 (on-reserve)
Location	Remote, island	Remote, mountainous	Remote, lakeside	Remote, river and lakeside
Climate	Temperate	Temperate, cold	Sub-Arctic	Humid continental
Governance	Haida Nation	Gitxsan Government Commission	Meadow Lake Tribal Council	Southeast Resource Development Council Corporation
Schools	4 elementary, 2 high school	7 elementary, 1 high school	1 school: K – Grade 12	1 school: nursery – grade 10
Partner Organisation	Secretariat of the Haida Nation	Storytellers' Foundation	Meadow Lake Tribal Council	Black River First Nation Chief and Council

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855 Table 2: Goals as Articulated by the Learning Circle Participants in Each Community*

Goal†	HG	HZ	MK	BR‡
Connection between Elders and youth	✓			✓
Curriculum development	✓	✓		✓
Funding/sustainability	✓	✓	✓	
Increased access to healthy foods	✓	✓	✓	
Increased access to traditional foods; learning traditional skills	✓	✓	✓	✓
Partnership development in-community	✓		✓	

856 * Bold check marks indicate goals that while not mentioned explicitly in community documents, were seen in the
857 activities of the work in that community. †Goals have been paraphrased according to the theme of the goal. ‡While BR
858 never held a formal LC meeting, a range of community members worked with the Learning Circle Evaluation Facilitator
859 on these implicit goals.

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Table 3: LC:LHF2S Communities Compared by Activity and Theme*

	Haida Gwaii	Hazelton/Upper Skeena	Ministikwan Lake	Black River
Activities†				
Gardening	✓	✓	✓	✓
Hospital food	✓			
Local food pantry	✓			
School food	✓		✓	✓
Skills development and capacity building	✓	✓	✓	✓
Youth leadership	✓	✓	✓	✓
Themes				
Access to healthy food; local food	✓	✓	✓	✓
Building partnerships and community togetherness	✓	✓	✓	
Connections between youth and Elders	✓	✓	✓	✓
Knowledge exchange; developing skills	✓	✓	✓	
Local champions	✓	✓	✓	
Traditional food, knowledge and practices	✓	✓	✓	✓
Transitioning to Indigenous leadership	✓			

*Bold check marks are used to indicate activities and themes that, while not mentioned explicitly in community codes, were seen as throughout the work in that specific community. †Activity and theme titles have been paraphrased from those presented in each individual case study.

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